

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and deployed **Java Spring Boot** microservices for backend systems, implementing **RESTful APIs** to facilitate cross-service data exchange.
- Engineered and maintained backend services using **Hibernate** and **JPA**, managing database interactions and ensuring data integrity across relational databases.
- Authored unit and integration tests using **JUnit** and **Mockito** for Java applications, achieving over 85% code coverage for critical modules.
- Integrated and operationalized **LLM** capabilities, including RAG-style retrieval and summarization, into enterprise applications for enhanced user experiences.
- Built and evolved **IoT** integrations for telemetry ingestion and event processing, focusing on device status monitoring and system integrations for smart building environments.
- Designed and implemented scalable cloud-native applications on **AWS**, leveraging services like EC2, S3, and Lambda to support distributed systems.
- Containerized applications using **Docker** and managed deployments in **Kubernetes** clusters, configuring Helm charts for efficient and repeatable deployments.

SKILLS

Languages – Java, Python, SQL, R, TypeScript

Frameworks & Libraries – Spring Boot, Hibernate, JPA, JUnit, Mockito, scikit-learn, pandas, NumPy, LangChain, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Helm, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle, NoSQL

AI & IoT – LLM, Generative AI, RAG, IoT Architectures, Telemetry Ingestion, Device Management, Smart Building Systems

Tools & Platforms – Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau

PROJECTS

AI-Powered Document Summarization Service (HCL Technologies)

- Developed a **Python**-based service utilizing **LangChain** and a pre-trained **LLM** to perform automated summarization of lengthy enterprise documents.
- Integrated a **RAG** pattern to enable retrieval of relevant context from a vector database, improving summarization accuracy by 18% on test datasets.
- Deployed the service as a **FastAPI** application within a **Docker** container, accessible via **REST APIs** for integration into existing workflows.

IoT Device Telemetry Ingestion Pipeline (HCL Technologies)

- Engineered a scalable **AWS Lambda**-based pipeline to ingest and process telemetry data from a fleet of 5,000+ IoT devices.
- Implemented data validation and transformation logic using **Python** and **pandas**, ensuring data quality before storage in **S3**.
- Configured **CloudWatch** alarms for anomaly detection in data ingestion rates, reducing critical data loss incidents by approximately 22%.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)